

Solutions to First Problem Sheet

on

Analytical Mechanics

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Topic: Newtonian Mechanics

"Millions saw the apple fall, but Newton was the one who asked why."

-Bernard Baruch

Problem 1

Prove Proposition 1.1, 1.3 and 1.4.

Solution 1

1.1) Use

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

and $\sin^2 \theta = 1 - \cos^2 \theta$. The unit vector is

$$n = \frac{u \times v}{\|u \times v\|}$$

so that straightforward computation yields the desired result.

1.3) Let $\mathcal{I}'$ be a frame moving in uniform motion (no acceleration) with respect to another inertial reference frame $\mathcal{I}$, then

$$\mathbf{r}_{P/O} = \mathbf{r}_{P/O'} + \mathbf{R}_{O'/O},$$

which on taking the double derivative respect to $\mathcal{I}$ yields the desired result.

1.4) We use the well-known fact that $SO(3)$ is a group to justify some properties.

i) Closure: taking the composition of two arbitrary g_1, g_2 we get the new element

$$g_1 \circ g_2 = (R_1 R_2, R_2 \mathbf{v}_1 + \mathbf{v}_2, R_2 \mathbf{x}_0^1 + \mathbf{x}_0^2, t_0^1 + t_0^2)$$

which is still in $\text{Gal}(3)$ because $SO(3)$ is a group, $R_2 \in \text{End}(\mathbb{R}^3)$, $\mathbb{R}^3$ is a group under vector addition (defined componentwise) and $\mathbb{R}$ is a group under the usual addition, respectively.

ii) Associativity: this property is endowed by the underlying groups.

iii) Existence of an identity: the element $g = (I, 0, 0, 0)$ is clearly in the set, I being the identity of $SO(3)$.

iv) Existence of inverses: given any element $g = (R, \mathbf{v}, \mathbf{v}, \mathbf{x}_0, t_0)$, the inverse can be shown to be $g^{-1} = (R^{-1}, -R^{-1}\mathbf{v}, -R^{-1}\mathbf{x}_0, -t_0)$, with the existence of each element composing g^{-1} being endowed primarily from $SO(3)$.



Problem 2

Consider the transformation for rotating frames of reference $\mathbf{r}'(t) = \mathbf{R}(t)\mathbf{r}(t)$, $\mathbf{R}(t) \in SO(3)$.

- i) Prove that $\dot{\mathbf{R}}\mathbf{R}^T$ is antisymmetric and then show that any antisymmetric matrix $\Omega \in \mathcal{M}^{3 \times 3}$ can be written as $(\Omega)_{ij} = \varepsilon_{ijk}\omega_k$ for some $\boldsymbol{\omega} \in \mathbb{R}^3$.

Next, let $\mathcal{I} = \{O, e_1, e_2, e_3\}$ be an inertial reference frame, then consider the non-inertial reference frame $\mathcal{N} = \{O, e'_1, e'_2, e'_3\}$ with $e'_i = \mathbf{R}e_i$. Using direct differentiation, deduce:

- ii) The so-called **transport equation**

$$\frac{\mathcal{N}d}{dt}\mathbf{r}' = \frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times \mathbf{r}'.$$

- iii) The analogue of Newton's second law of motion for rotating (noninertial) frames

$$\mathcal{N}\mathbf{F} = \mathcal{I}\mathbf{F} + 2\boldsymbol{\omega} \times \mathcal{I}\mathbf{v}' + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}').$$

Further, identify each term in the latter expression.

Solution 2

- i) From the definition of $\mathbf{R}$ we get $\mathbf{R}\mathbf{R}^T = \mathbf{I}$ so $\dot{\mathbf{R}}\mathbf{R}^T + \mathbf{R}\dot{\mathbf{R}}^T = 0$, that is, $\dot{\mathbf{R}}\mathbf{R}^T = -\mathbf{R}\dot{\mathbf{R}}^T$, and it is now clear that $\dot{\mathbf{R}}\mathbf{R}^T$ is antisymmetric. Any antisymmetric matrix Ω satisfies $(\Omega)_{ij} = -(\Omega)_{ji}$ from which $(\Omega)_{jj} = 0$ and thus $(\Omega)_{ij} = \varepsilon_{ijk}\omega_k$ (there is only three degrees of freedom in choosing the entries of an antisymmetric matrix). This also shows that we may write $\dot{\mathbf{R}}\mathbf{R}^T = \Omega$ and hence $\dot{\mathbf{R}} = \Omega\mathbf{R}$.
- ii) Direct differentiation of $\mathbf{r}'$ yields

$$\frac{\mathcal{I}d}{dt}\mathbf{r}' = \dot{\mathbf{R}}\mathbf{r} + \mathbf{R}\frac{\mathcal{I}d}{dt}\mathbf{r}$$

but notice that on writing $\mathbf{r} = r^i e_i$ we have that the second term in the previous equation is just $\mathbf{R}(\dot{r}^i e_i) = \dot{r}^i e'_i$, which is exactly the same as the derivative in frame $\mathcal{N}$ of $\mathbf{r}'$, thus

$$\frac{\mathcal{I}d}{dt}\mathbf{r}' = \Omega\mathbf{R}\mathbf{r} + \frac{\mathcal{N}d}{dt}\mathbf{r}' = -\boldsymbol{\omega} \times \mathbf{r}' + \frac{\mathcal{N}d}{dt}\mathbf{r}'$$

and the conclusion now readily follows.

- iii) Direct application of the transport equation to itself yields

$$\begin{aligned} \frac{\mathcal{N}d}{dt}\frac{\mathcal{N}d}{dt}\mathbf{r}' &= \frac{\mathcal{N}d}{dt}\left(\frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times \mathbf{r}'\right) = \frac{\mathcal{I}d}{dt}\frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times \frac{\mathcal{I}d}{dt}\mathbf{r}' + \frac{\mathcal{I}d}{dt}(\boldsymbol{\omega} \times \mathbf{r}') + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\ &= \frac{\mathcal{I}d}{dt}\frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times \frac{\mathcal{I}d}{dt}\mathbf{r}' + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \boldsymbol{\omega} \times \frac{\mathcal{I}d}{dt}\mathbf{r}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\ &= \frac{\mathcal{I}d}{dt}\frac{\mathcal{I}d}{dt}\mathbf{r}' + 2\boldsymbol{\omega} \times \frac{\mathcal{I}d}{dt}\mathbf{r}' + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \end{aligned}$$

so that on multiplying this expression through the mass m of a particle gives the desired result. The first term is the only real force, it may be called the "inertial force", the second term is known as the Coriolis force, the third one is sometimes called the Euler force and the last one is the well-known centrifugal force.

✱

Problem 3

Prove Theorem 1.7 (Conservation Theorem for the Angular Momentum of a Particle).

Solution 3

The result is an immediate consequence of

$$\frac{d}{dt} \mathcal{I} \mathbf{L}_{P/O} = \frac{d}{dt} \mathbf{r}_{P/O} \times \mathcal{I} \mathbf{p}_{P/O} + \mathbf{r}_{P/O} \times \frac{d}{dt} \mathcal{I} \mathbf{p}_{P/O} = \mathbf{N}_{P/O}.$$



Problem 4

Prove that if U is a simply connected open subset of $\mathbb{R}^n$, then the following statements are equivalent:

- i) $\int_{\Gamma} \mathbf{F} = \int_{\Sigma} \mathbf{F}$ for all contours $\Gamma, \Sigma \subset U$ joining the same initial and final points.
- ii) $\int_D \mathbf{F} = 0$ for all closed contours $D \subset U$.
- iii) $\mathbf{F} = \nabla V$ for some $V : U \rightarrow \mathbb{R}$ at least class $C^1(U)$.

Moreover, show that for the case $n = 3$ we also have the equivalence with

- iv) $\nabla \times \mathbf{F} = 0$ on U .

Hint: for iv) you may want to look up Stoke's theorem from vector analysis.

Solution 4

iii) $\Rightarrow$ ii) If $\mathbf{F} = \nabla V$, then

$$\oint_{\gamma} \mathbf{F} = \oint_{\gamma} \frac{\partial V}{\partial x_i} e_i dx_j e_j = \oint_{\gamma} dV = V(b) - V(a)$$

where a, b are the endpoint of γ , which are the same since γ is a closed contour and thus the integral vanishes identically.

ii) $\Rightarrow$ i) Given any point $a \in U$, we can choose any other $b \in U$ and let Γ, Σ be two distinct paths connecting the points a, b but since U is connected, these paths lie entirely in U , so that let $\gamma = \Gamma \cup \Sigma^-$ be the union of the path Γ and the reverse path Σ^- , so that γ is closed and lies entirely on U , so

$$\oint_{\gamma} \mathbf{F} = \int_{\Gamma} \mathbf{F} + \int_{\Sigma^-} \mathbf{F} = \int_{\Gamma} \mathbf{F} - \int_{\Sigma} \mathbf{F} = 0$$

and thus the result now directly follows from applying the assumption that the closed integral identically vanishes.

i) $\Rightarrow$ iii) Using the insight from calculus, we propose that for any $x_0 \in U$ fixed, the function

$$V(x) = \int_{\delta} \mathbf{F}$$

where δ is any path connecting x_0 to x , satisfies $\mathbf{F} = \nabla V$, to prove so, we show that $\partial_i V = F_i$, so consider the difference

$$\left| \frac{V(x + h_i) - V(x)}{h} - F_i \right| = \left| \frac{1}{h} \left(\int_{x_0}^{x+h_i} \mathbf{F} - \int_{x_0}^x \mathbf{F} \right) - F_i \right|$$

with $h_i = h e_i$ and hence using that $h = \int_x^{x+h_i} d\mu$, we have

$$\left| \frac{1}{h} \left(\int_{x_0}^{x+h_i} \mathbf{F} - \int_{x_0}^x \mathbf{F} \right) - F_i \right| = \frac{1}{|h|} \left| \int_x^{x+h_i} \mathbf{F} \right|$$



Problem 5

Show that every central force is conservative. In particular, for Newton's gravitational force, find the potential $V(r)$.

Solution 5

A central force is characterised by its dependence on the distance between particles only, meaning that we can write (at least for relative motion) $\mathbf{F} = f(r)\hat{\mathbf{r}}$ with f a differentiable function of $r = \sqrt{x^2 + y^2 + z^2}$. In Cartesian coordinates the computation is a little clumsy, so let us use spherical coordinates (r, θ, ϕ) , in which the operator $\nabla \times$ takes the form

$$\nabla \times = \frac{1}{r^2 \sin \theta} \left[\hat{r}(\partial_\theta(r \sin \theta \pi^\phi) - \partial_\phi(r \pi^\theta)) + r \hat{\theta}(\partial_\phi \pi^r - \partial_r(r \sin \theta \pi^\phi)) + r \sin \theta \hat{\phi}(\partial_r(r \pi^\theta) - \partial_\theta \pi^r) \right]$$

with π^i the canonical projection in the i th coordinate. It is now evident that for a central force $\mathbf{F}$ we have $\pi^\phi \mathbf{F} = \pi^\theta \mathbf{F} = 0$ and $\pi^r \mathbf{F} = f(r)$ so

$$\nabla \times \mathbf{F} = \frac{1}{r^2 \sin \theta} \left[r \hat{\theta} \partial_\phi f(r) - r \sin \theta \hat{\phi} \partial_\theta f(r) \right] = 0$$

which by the previous problem is equivalent to being conservative. In particular, for Newton's gravitational force, it can be seen from inspection that

$$\mathbf{F} = -G \frac{m_1 m_2}{r^3} \mathbf{r} = -\nabla V, \quad V(r) = -G \frac{m_1 m_2}{r}.$$



Problem 6

If a projectile is fired from the origin of the coordinate system with an initial velocity v_0 and in a direction making an angle α with the horizontal, calculate the time required for the projectile to cross a line passing through the origin and making an angle $\beta < \alpha$ with the horizontal (neglect the effects of drag forces).

Solution 6

Neglecting air resistance, the only force acting on the projectile is gravity. Hence, Newton's second law gives

$$m\ddot{\mathbf{r}} = -mg\hat{\mathbf{j}},$$

If the projectile is fired with initial speed v_0 at an angle α with the horizontal, then assume $x(0) = 0$, $y(0) = 0$ and $\dot{x}(0) = v_0 \cos \alpha$, $\dot{y}(0) = v_0 \sin \alpha$. Integrating the equations of motion, we obtain

$$x(t) = v_0 \cos \alpha t,$$

and

$$y(t) = v_0 \sin \alpha t - \frac{1}{2}gt^2.$$

The line through the origin making an angle β with the horizontal has equation $y = x \tan \beta$, therefore, the projectile crosses this line when

$$v_0 \sin \alpha t - \frac{1}{2}gt^2 = v_0 \cos \alpha t \tan \beta.$$

Factoring out t gives

$$t \left(v_0 \sin \alpha - v_0 \cos \alpha \tan \beta - \frac{1}{2}gt \right) = 0,$$

where the solution $t = 0$ corresponds to the launch point and is discarded. Thus,

$$t = \frac{2v_0}{g} (\sin \alpha - \cos \alpha \tan \beta).$$

Using the identity

$$\sin \alpha - \cos \alpha \tan \beta = \frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \beta} = \frac{\sin(\alpha - \beta)}{\cos \beta},$$

we conclude that the required time is

$$t = \frac{2v_0 \sin(\alpha - \beta)}{g \cos \beta}.$$

Since $\beta < \alpha$, we have $\sin(\alpha - \beta) > 0$, and therefore the obtained time is positive, as expected. $\star$

Problem 7

Consider a particle undergoing horizontal motion in a medium having a drag force satisfying Stoke's law. Find the position and velocity of the particle. Show that the particle eventually stops (as expected) and find the total range. If the particle started with a velocity $\mathbf{v}_0 = v_0 \hat{\mathbf{x}}$, find the work done by the drag force on the trajectory of the particle.

Solution 7

Suppose the particle moves along the x -axis and is subjected only to a drag force satisfying Stoke's law, $\mathbf{F}_d = -k\mathbf{v}$, where $k > 0$ is the drag coefficient. By Newton's second law,

$$m \frac{dv}{dt} = -kv.$$

This is a separable differential equation, so

$$\frac{dv}{v} = -\frac{k}{m} dt,$$

integrating both sides gives,

$$\ln |v| = -\frac{k}{m} t + C,$$

by the initial condition $v(0) = v_0$, we obtain

$$v(t) = v_0 e^{-kt/m}.$$

Since $v = dx/dt$, the position is given by

$$\frac{dx}{dt} = v_0 e^{-kt/m},$$

integrating once more and imposing $x(0) = 0$,

$$x(t) = \frac{mv_0}{k} (1 - e^{-kt/m}).$$

Observe that

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} v_0 e^{-kt/m} = 0,$$

so the particle eventually comes to rest. Likewise,

$$\lim_{t \rightarrow \infty} x(t) = \frac{mv_0}{k},$$

Hence, the total range traveled by the particle is

$$R = \frac{mv_0}{k},$$

and finally, the work done by the drag force is

$$W_d = \int_C \mathbf{F}_d \cdot d\mathbf{r}.$$

Since the motion is one-dimensional,

$$\begin{aligned} W_d &= - \int_0^R kv \, dx. \\ &= -k \int_0^\infty v^2 \, dt. \end{aligned}$$

Substituting the expression for the velocity,

$$W_d = -kv_0^2 \int_0^\infty e^{-2kt/m} \, dt,$$

since

$$\int_0^\infty e^{-2kt/m} \, dt = \frac{m}{2k},$$

we conclude that

$$W_d = -\frac{1}{2}mv_0^2.$$

This agrees with the work-energy theorem, since the particle ultimately comes to rest and all of its initial kinetic energy is dissipated by the drag force.



Problem 8

Consider a particle vertically falling in a medium satisfying Stoke's law. Find the position and velocity of the particle. Furthermore, compute

- i) $\lim_{t \rightarrow \infty} \mathbf{v}(t),$
- ii) $\lim_{t \rightarrow \infty} \mathbf{r}(t),$
- iii) W_d (the work done by the drag force)

and discuss whether the results are physically consistent.

Solution 8

Let the positive y -axis point downward so that gravity acts in the positive direction. The forces acting on the particle are gravity and the Stoke's drag force,

$$F_g = mg, \quad F_d = -kv,$$

where $k > 0$ is the drag coefficient. Newton's second law becomes

$$m \frac{dv}{dt} = mg - kv.$$

Defining

$$\xi = \frac{m}{k},$$

the equation takes the form

$$\frac{dv}{dt} + \frac{1}{\xi}v = g.$$

Since the particle starts from rest, $v(0) = 0$, the solution is

$$v(t) = g\xi \left(1 - e^{-t/\xi}\right) = \frac{mg}{k} \left(1 - e^{-kt/m}\right),$$

integrating the velocity and using $r(0) = 0$,

$$r(t) = \int_0^t v(s) ds = g\tau t - g\tau^2 \left(1 - e^{-t/\tau}\right),$$

or equivalently,

$$r(t) = \frac{mg}{k}t - \frac{m^2g}{k^2} \left(1 - e^{-kt/m}\right).$$

The terminal velocity is therefore

$$\lim_{t \rightarrow \infty} v(t) = g\tau = \frac{mg}{k}$$

and also,

$$r(t) = g\xi t - g\xi^2 \left(1 - e^{-t/\xi}\right),$$

so that

$$\lim_{t \rightarrow \infty} r(t) = +\infty.$$

This is expected, since the particle continues falling indefinitely while approaching the constant terminal velocity, the work done by the drag force is

$$W_d = \int_0^\infty F_d dr = \int_0^\infty (-kv)v dt = -k \int_0^\infty v^2(t) dt,$$

substituting the expression for the velocity,

$$W_d = -k \left(\frac{mg}{k}\right)^2 \int_0^\infty \left(1 - e^{-kt/m}\right)^2 dt.$$

Since

$$\left(1 - e^{-kt/m}\right)^2 \rightarrow 1 \quad (t \rightarrow \infty),$$

the integral diverges, and therefore

$$W_d = -\infty.$$

This result is physically consistent. Although the drag force remains bounded, $F_d \rightarrow -mg$, the particle travels an unbounded distance while moving with the nonzero terminal velocity mg/k . Consequently, the drag force performs an unbounded amount of negative work over an infinite time interval.



Problem 9

Consider the motion of a projectile fired with a velocity $\mathbf{v}_0$. Neglect the effects of the air resistance and thus find its position, velocity and range. If air resistance is taken into consideration (assume that Stoke's law holds), calculate the decrease in range by using the **perturbation method**.

Solution 9

Neglecting air resistance, the motion of the projectile is governed by

$$m\ddot{\mathbf{r}} = -mg\hat{\mathbf{j}},$$

Assuming the projectile is launched from the origin with speed v_0 at an angle θ above the horizontal, the initial conditions are $x(0) = 0, y(0) = 0$ and $\dot{x}(0) = v_0 \cos \theta, \dot{y}(0) = v_0 \sin \theta$. Integrating the equations of motion gives the velocity

$$v_x(t) = v_0 \cos \theta, \quad v_y(t) = v_0 \sin \theta - gt,$$

and the position

$$x(t) = v_0 \cos \theta t, \\ y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2.$$

The projectile reaches the ground when $y(t) = 0$. Besides the trivial solution $t = 0$, we obtain

$$t_f = \frac{2v_0 \sin \theta}{g},$$

therefore, the range is

$$R_0 = x(t_f) = \frac{v_0^2}{g} \sin(2\theta).$$

Now suppose that the projectile experiences a drag force, then Newton's second law becomes

$$m\dot{\mathbf{v}} = -mg\hat{\mathbf{j}} - k\mathbf{v},$$

introducing the small parameter

$$\varepsilon = \frac{k}{m},$$

the equations of motion are

$$\dot{v}_x = -\varepsilon v_x, \quad \dot{v}_y = -g - \varepsilon v_y.$$

whose solutions are

$$v_x(t) = v_0 \cos \theta e^{-\varepsilon t}, \\ v_y(t) = \left(v_0 \sin \theta + \frac{g}{\varepsilon}\right) e^{-\varepsilon t} - \frac{g}{\varepsilon}.$$

Assuming $\varepsilon \ll 1$, we use the first-order approximation $e^{-\varepsilon t} \approx 1 - \varepsilon t$. Hence,

$$v_x(t) \approx v_0 \cos \theta (1 - \varepsilon t),$$

and

$$v_y(t) \approx v_0 \sin \theta - gt - \varepsilon v_0 \sin \theta t + \frac{1}{2}\varepsilon gt^2.$$

Integrating once more yields

$$x(t) \approx v_0 \cos \theta \left(t - \frac{1}{2}\varepsilon t^2\right),$$

and

$$y(t) \approx v_0 \sin \theta t - \frac{1}{2}gt^2 - \frac{1}{2}\varepsilon v_0 \sin \theta t^2 + \frac{1}{6}\varepsilon gt^3.$$

Let the flight time be written as

$$t_f = t_0 + \varepsilon t_1,$$

where

$$t_0 = \frac{2v_0 \sin \theta}{g}$$

is the flight time without drag. Substituting this expansion into $y(t_f) = 0$ and retaining only first-order terms in ε , one finds

$$t_1 = -\frac{2v_0^2 \sin^2 \theta}{3g^2}.$$

Consequently,

$$t_f \approx \frac{2v_0 \sin \theta}{g} - \frac{2kv_0^2 \sin^2 \theta}{3mg^2},$$

the corresponding range is

$$R = x(t_f) \approx R_0 - \frac{4kv_0^3}{3mg^2} \sin^2 \theta \cos \theta.$$

Therefore, the decrease in range due to Stoke's drag is

$$\Delta R = R_0 - R = \frac{4kv_0^3}{3mg^2} \sin^2 \theta \cos \theta.$$

As expected, $\Delta R > 0$, so the presence of air resistance decreases the distance traveled by the projectile. ✱

Problem 10

Consider a particle of mass m whose motion starts from rest in a constant gravitational field. If a drag force satisfying Stoke's law is encountered, show that the distance s the particle falls in accelerating from v_0 to v_1 is given by

$$\ln \left(\frac{g - kv_1}{g - kv_0} \right) + (v_0 - v_1).$$

Furthermore, prove that if a drag force proportional to the square of the velocity is encountered, then s is given by

$$d = \frac{1}{2k} \ln \frac{g - kv_0^2}{g - kv_1^2}$$

where k is the proportionality factor.

Solution 10

Let the positive x -axis point downward, so that gravity acts in the positive direction, and write the drag force per unit mass as $-kv$ with $k > 0$. Newton's second law then reads

$$\frac{dv}{dt} = g - kv.$$

Since we are interested in the distance travelled rather than in the elapsed time, it is convenient to eliminate t by means of the chain rule,

$$\frac{dv}{dt} = \frac{dv}{ds} \frac{ds}{dt} = v \frac{dv}{ds},$$

so that the equation of motion becomes

$$v \frac{dv}{ds} = g - kv.$$

Separating variables and integrating between the speeds v_0 and v_1 , the distance fallen is

$$s = \int_{v_0}^{v_1} \frac{v dv}{g - kv},$$

and using the partial fraction decomposition

$$\frac{v}{g - kv} = -\frac{1}{k} + \frac{g}{k} \frac{1}{g - kv},$$

we obtain

$$s = -\frac{1}{k} \int_{v_0}^{v_1} dv + \frac{g}{k} \int_{v_0}^{v_1} \frac{dv}{g - kv} = -\frac{v_1 - v_0}{k} - \frac{g}{k^2} \left[\ln(g - kv) \right]_{v_0}^{v_1}.$$

Hence

$$s = \frac{g}{k^2} \ln \left(\frac{g - kv_0}{g - kv_1} \right) + \frac{1}{k} (v_0 - v_1),$$

which is the asserted expression once the dimensional constants are restored, since $\ln[(g - kv_0)/(g - kv_1)] = -\ln[(g - kv_1)/(g - kv_0)]$. Observe that the result makes sense only for speeds below the terminal one, $v_t = g/k$, for otherwise the logarithm is undefined; this is no restriction, because a particle released from rest approaches v_t asymptotically and never attains it. Moreover, the integrand $v/(g - kv)$ is positive on $0 \leq v < v_t$, so that $s > 0$ whenever $v_1 > v_0$, as expected. In particular, if the particle is timed from rest, $v_0 = 0$ and

$$s = \frac{g}{k^2} \ln \left(\frac{g}{g - kv_1} \right) - \frac{v_1}{k}.$$

Suppose now that the retarding force is proportional to the square of the speed, so that the equation of motion is

$$\frac{dv}{dt} = g - kv^2,$$

where k is the new proportionality factor. Proceeding exactly as before,

$$v \frac{dv}{ds} = g - kv^2,$$

whence

$$d = \int_{v_0}^{v_1} \frac{v dv}{g - kv^2},$$

and since

$$\frac{d}{dv} (g - kv^2) = -2kv,$$

the integral is immediate,

$$d = -\frac{1}{2k} \left[\ln(g - kv^2) \right]_{v_0}^{v_1} = \frac{1}{2k} \ln \left(\frac{g - kv_0^2}{g - kv_1^2} \right),$$

which is the desired result. As in the linear case, the expression is meaningful only for speeds below the terminal one, now $v_t = \sqrt{g/k}$, and because $v_1 > v_0$ the argument of the logarithm exceeds unity, so that $d > 0$, as expected. ✱

Problem 11

A particle moves in a medium under the influence of a retarding force equal to $mk(v^3 + a^2v)$, where k and a are constants. Show that for any value of the initial speed the particle will never move a distance greater than $\pi/2ka$ and that the particle comes to rest only for $t \rightarrow \infty$.

Solution 11

The retarding force is

$$\mathbf{F} = -mk(v^3 + a^2v) \hat{\mathbf{v}},$$

where $k, a > 0$. Since the force always opposes the motion, Newton's second law gives

$$\frac{dv}{dt} = -kv(v^2 + a^2),$$

rewriting the equation gives

$$\frac{dv}{v(v^2 + a^2)} = -k dt$$

and using the partial fraction decomposition

$$\frac{1}{v(v^2 + a^2)} = \frac{1}{a^2} \left(\frac{1}{v} - \frac{v}{v^2 + a^2} \right),$$

we obtain

$$\frac{1}{a^2} \left(\ln |v| - \frac{1}{2} \ln(v^2 + a^2) \right) = -kt + C.$$

Hence,

$$\ln \left(\frac{v}{\sqrt{v^2 + a^2}} \right) = -a^2 kt + C',$$

as $t \rightarrow \infty$, the right-hand side tends to $-\infty$, so

$$\frac{v}{\sqrt{v^2 + a^2}} \rightarrow 0,$$

which implies $\lim_{t \rightarrow \infty} v(t) = 0$. Therefore, the particle comes to rest only asymptotically, that is, only as $t \rightarrow \infty$. Now to determine the distance traveled, observe that $v = dx/dt$, then by the chain rule and, since $v > 0$ during the motion,

$$\frac{dv}{dx} = -k(v^2 + a^2).$$

Thus,

$$dx = -\frac{dv}{k(v^2 + a^2)}.$$

If the initial speed is v_0 , the total distance traveled before coming to rest is

$$\begin{aligned} R &= \int_0^{v_0} \frac{dv}{k(v^2 + a^2)} \\ &= \frac{1}{ka} \arctan\left(\frac{v_0}{a}\right). \end{aligned}$$

Because $\arctan\left(\frac{v_0}{a}\right) < \frac{\pi}{2}$, for every finite value of v_0 , it follows that $R < \frac{\pi}{2ka}$.

Hence, regardless of the initial speed, the particle can never travel a distance greater than $\frac{\pi}{2ka}$, while its velocity approaches zero only in the limit $t \rightarrow \infty$.



Problem 12

Consider a particle moving in the region $x > 0$ under the influence of the potential

$$V(x) = V_0 \left(\frac{a}{x} + \frac{x}{a} \right)$$

where $V_0 = 1J$ and $a = 2m$. Plot the potential, find the equilibrium points, and determine whether they are maxima or minima.

Solution 12

The potential energy is

$$V(x) = V_0 \left(\frac{a}{x} + \frac{x}{a} \right),$$

where $V_0 = 1 \text{ J}$ and $a = 2 \text{ m}$. To sketch the potential, we first analyze its qualitative behavior. As $x \rightarrow 0^+$, $\frac{a}{x} \rightarrow \infty$, and therefore $V(x) \rightarrow \infty$. Likewise, as $x \rightarrow \infty$, $\frac{x}{a} \rightarrow \infty$, so that $V(x) \rightarrow \infty$. Hence, the potential diverges at both ends of the interval $(0, \infty)$ and must possess at least one minimum. The equilibrium points satisfy

$$F(x) = -\frac{dV}{dx} = 0.$$

Since

$$\frac{dV}{dx} = V_0 \left(-\frac{a}{x^2} + \frac{1}{a} \right),$$

the equilibrium condition becomes

$$-\frac{a}{x^2} + \frac{1}{a} = 0,$$

multiplying by ax^2 gives $x^2 = a^2$, and because the motion is restricted to the region $x > 0$, the only equilibrium point is $x = a$. To determine its nature, we compute the second derivative

$$\frac{d^2V}{dx^2} = \frac{2aV_0}{x^3},$$

and by evaluating at the equilibrium point gives $\frac{d^2V}{dx^2} > 0$. Therefore, the equilibrium at $x = a$ is a local minimum and is consequently stable. Finally, using the given numerical values,

$$V(x) = \frac{2}{x} + \frac{x}{2},$$

whose minimum value is

$$V(2) = \frac{2}{2} + \frac{2}{2} = 2 \text{ J}.$$

Thus, the graph of the potential starts at $+\infty$ as $x \rightarrow 0^+$, decreases until reaching its unique minimum at $(2, 2)$, and then increases monotonically to $+\infty$ as $x \rightarrow \infty$.



Problem 13

Write the criteria for determining whether an equilibrium is stable or unstable when all derivatives up through order n vanish. Prove your assertion.

Solution 13

Let x_0 be an equilibrium point of the potential V , so that

$$V'(x_0) = 0.$$

Suppose further that

$$V'(x_0) = V''(x_0) = \cdots = V^{(n)}(x_0) = 0,$$

while $V^{(n+1)}(x_0) \neq 0$. To determine the nature of the equilibrium, we expand V about x_0 using Taylor's theorem:

$$V(x) = V(x_0) + \frac{V^{(n+1)}(x_0)}{(n+1)!} (x - x_0)^{n+1} + o((x - x_0)^{n+1}).$$

There are two cases. If $n+1$ is odd, then the first nonzero term changes sign as x passes through x_0 . Consequently, $V(x)$ is greater than $V(x_0)$ on one side of the equilibrium and smaller on the other. Thus, x_0 is neither a local maximum nor a local minimum, but a saddle point, and the equilibrium is unstable. If $n+1$ is even, then $(x-x_0)^{n+1} \geq 0$, so the sign of the first nonzero coefficient determines the behavior of the potential. If

$$V^{(n+1)}(x_0) > 0,$$

then

$$V(x) > V(x_0)$$

for all x sufficiently close to x_0 , showing that x_0 is a local minimum and the equilibrium is stable. On the other hand, if

$$V^{(n+1)}(x_0) < 0,$$

then

$$V(x) < V(x_0)$$

near x_0 , so x_0 is a local maximum and the equilibrium is unstable. Therefore, the criterion is as follows:

- a) If the first nonzero derivative after V' is of odd order, the equilibrium is unstable.
- b) If the first nonzero derivative is of even order and positive, the equilibrium is stable.
- c) If the first nonzero derivative is of even order and negative, the equilibrium is unstable.

This generalizes the familiar second derivative test, which corresponds to the case $n = 1$.



Problem 14

Consider a particle of mass m under the influence of the potential

$$V(\mathbf{r}) = V_0 \left[\left(\frac{x}{a} \right)^2 + \left(\frac{y}{a} \right)^2 - \frac{z/a}{1 + (z/a)^2} \right]$$

with V_0, a positive constants.

- i) Find the force $\mathbf{F}$ acting on the particle.
- ii) Find all the equilibrium points and determine whether they are stable, unstable or metastable (saddle point).
- iii) Sketch or describe distinct level surfaces of V .

Solution 14

The potential is

$$V(x, y, z) = V_0 \left[\left(\frac{x}{a} \right)^2 + \left(\frac{y}{a} \right)^2 - \frac{z/a}{1 + (z/a)^2} \right],$$

where $V_0, a > 0$. Since the force is given by $\mathbf{F} = -\nabla V$, we first compute the partial derivatives

$$\frac{\partial V}{\partial x} = \frac{2V_0 x}{a^2}, \quad \frac{\partial V}{\partial y} = \frac{2V_0 y}{a^2}.$$

For the z -component, let $u = z/a$. Then

$$V(x, y, z) = V_0 \left[\left(\frac{x}{a} \right)^2 + \left(\frac{y}{a} \right)^2 - \frac{u}{1 + u^2} \right].$$

Also, from

$$\frac{d}{du} \left(\frac{u}{1 + u^2} \right) = \frac{1 - u^2}{(1 + u^2)^2},$$

the chain rule gives

$$\frac{\partial V}{\partial z} = -\frac{V_0}{a} \frac{1 - (z/a)^2}{[1 + (z/a)^2]^2},$$

then,

$$\mathbf{F} = -\nabla V = -\frac{2V_0x}{a^2}\hat{\mathbf{i}} - \frac{2V_0y}{a^2}\hat{\mathbf{j}} + \frac{V_0}{a} \frac{1 - (z/a)^2}{[1 + (z/a)^2]^2}\hat{\mathbf{k}}.$$

To determine the equilibrium points, we solve $\mathbf{F} = \mathbf{0}$. The first two equations immediately give $x = 0$ and $y = 0$ and the third becomes

$$1 - \left(\frac{z}{a}\right)^2 = 0,$$

whose solutions are $z = \pm a$. Hence, the equilibrium points are $(0, 0, a)$, $(0, 0, -a)$. To classify them, we compute the Hessian of the potential. But notice that

$$\frac{\partial^2 V}{\partial x^2} = \frac{\partial^2 V}{\partial y^2} = v \frac{2V_0}{a^2} > 0,$$

so it remains only to study the z -direction. Differentiating once more we get,

$$\frac{\partial^2 V}{\partial z^2} = \frac{2V_0z(3 - (z/a)^2)}{a^4[1 + (z/a)^2]^3}$$

and evaluating at the equilibrium points,

$$\frac{\partial^2 V}{\partial z^2}(0, 0, a) = \frac{V_0}{4a^2} > 0,$$

while

$$\frac{\partial^2 V}{\partial z^2}(0, 0, -a) = -\frac{V_0}{4a^2} < 0.$$

Thus, the Hessian at $(0, 0, a)$ is positive definite, so this point is a local minimum and therefore a stable equilibrium. At $(0, 0, -a)$ the second derivatives with respect to x and y are positive, whereas the second derivative with respect to z is negative. Consequently, the Hessian is indefinite, and $(0, 0, -a)$ is a saddle point. Finally, the level surfaces are determined by

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{a}\right)^2 - \frac{z/a}{1 + (z/a)^2} = \frac{C}{V_0},$$

where C is a constant. As $|z| \rightarrow \infty$, the last term tends to zero, so the level surfaces approach circular cylinders

$$x^2 + y^2 = \frac{a^2 C}{V_0}.$$

Near $z = a$, the potential has a local minimum, producing a potential well, whereas near $z = -a$ it has a saddle shape.

